

Simple programming using CASE and LOOP

Program 1:

Write a function that uses CASE statement where if monthly_value is equal to or less than 4000, then income_level will be set to 'Low Income'. If monthly_value is equal to or less than 5000, then income_level will be set to 'Avg Income'. Otherwise, income_level will be set to 'High Income'.

```
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> create database collegedb;
Query OK, 1 row affected (0.08 sec)

mysql> use collegedb;
Database changed
mysql> delimiter //
mysql> create function income_level(monthly_value int) returns varchar(20) deterministic begin declare income_level varchar(20); case when monthly_value <= 4000 then set income_level = 'Low Income'; when monthly_value <= 5000 then set income_level = 'Avg Income'; else set income_level = 'High Income'; end case; return income_level; end //
Query OK, 0 rows affected (0.16 sec)

mysql> delimiter ;
mysql> select income_level(3000);
+-----+
| income_level(3000) |
+-----+
| Low Income         |
+-----+
1 row in set (0.08 sec)

mysql> select income_level(4500);
+-----+
| income_level(4500) |
+-----+
| Avg Income         |
+-----+
1 row in set (0.05 sec)

mysql> select income_level(7000);
+-----+
| income_level(7000) |
+-----+
| High Income        |
+-----+
1 row in set (0.00 sec)

mysql> select income_level(3000) as '3000', income_level(4500) as '4500', income_level(7000) as '7000';
+-----+-----+-----+
| 3000 | 4500 | 7000 |
+-----+-----+-----+
| Low Income | Avg Income | High Income |
+-----+-----+-----+
1 row in set (0.04 sec)
```

Program 2:

Write a function that will use ITERATE statement which would cause the loop to repeat while income is less than 4000. Once income is greater than or equal to 4000, would terminate the LOOP.

```
Type 'help;' or '\h' for help. Type 'c' to clear the current input statement.
mysql> create database collegedb;
Query OK, 1 row affected (0.06 sec)

mysql> use collegedb;
Database changed
mysql> delimiter //
mysql> create function income_loop() returns int deterministic begin declare income int default 1000; myloop: loop set income = income + 1000; if income >=
4000 then leave myloop; end if; iterate myloop; end loop; return income; end //
Query OK, 0 rows affected (0.07 sec)

mysql> delimiter ;
mysql> select income_loop() as Final_Income;
+-----+
| Final_Income |
+-----+
|          4000 |
+-----+
1 row in set (0.07 sec)
```